In [ ]:
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# PCA of pcb congeners in stagnant and flowing water sediment (Dec to Feb)
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In [1]:
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import matplotlib.pyplot as plt
import numpy as np
import pandas as pd
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler

# 1. Define PCB names/congeners
names = [
    "PCB-8",
    "PCB-18",
    "PCB-28",
    "PCB-44",
    "PCB-52",
    "PCB-66",
    "PCB-77",
    "PCB-81",
    "PCB-101",
    "PCB-105",
    "PCB-114",
    "PCB-118",
    "PCB-123",
    "PCB-126",
    "PCB-128",
    "PCB-138",
    "PCB-153",
    "PCB-156",
    "PCB-157",
    "PCB-167",
    "PCB-169",
    "PCB-170",
    "PCB-180",
    "PCB-187",
    "PCB-189",
    "PCB-195",
    "PCB-206",
    "PCB-209",
]

# 2. Define datasets for Stagnant and Flowing water
stagnant = [
    [2.745, 0.57, 2.505],
    [10.83, 0.755, 1.08],
    [3.95, 0.73, 0.57],
    [19.425, 0.82, 1.005],
    [3.22, 0.515, 1.695],
    [10.77, 1.245, 0.515],
    [0.635, 1.31, 0.555],
    [8.25, 0.68, 0.54],
    [5.555, 0.55, 1.29],
    [10.64, 0.77, 0.625],
    [4.7, 0.635, 0.72],
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    [2.81, 1.535, 3.235],
    [4.085, 0.395, 2.23],
    [6.44, 1.525, 1.55],
    [4.46, 0.745, 1.965],
    [2.39, 0.405, 0.815],
    [9.005, 0.9, 1.57],
    [5.875, 1.01, 0.51],
    [4.255, 1.375, 1.985],
    [23.345, 0.51, 1.4],
    [4.16, 0.565, 1.09],
    [6.6, 0.71, 2.24],
    [7.57, 2.1, 2.005],
    [4.78, 0.85, 1.09],
    [7.92, 0.28, 0.775],
    [12.935, 0.52, 1.33],
    [7.34, 0.63, 1.015],
    [17.03, 0.715, 1.88],
]

flowing = [
    [3.35, 0.11, 0.62],
    [2.35, 0.31, 0.325],
    [1.35, 0.8, 0.82],
    [4.665, 0.82, 0.455],
    [3.68, 0.535, 0.275],
    [2.94, 0.34, 1.355],
    [0.315, 0.62, 0.225],
    [2.025, 0.35, 0.355],
    [0.665, 0.465, 0.795],
    [2.705, 0.43, 1.19],
    [0.92, 0.305, 0.825],
    [3.835, 0.365, 0.295],
    [0.6, 1.13, 1.23],
    [0.635, 0.31, 0.825],
    [0.71, 0.805, 0.41],
    [1.2, 0.94, 0.85],
    [3.145, 0.77, 1.62],
    [1.66, 0.605, 0.615],
    [1.245, 0.62, 0.735],
    [2.095, 0.53, 0.995],
    [1.71, 0.66, 2.365],
    [3.875, 0.72, 1.135],
    [2.96, 0.305, 1.0],
    [1.735, 0.53, 2.01],
    [4.985, 0.6, 0.62],
    [3.9, 0.465, 0.995],
    [3.135, 0.745, 0.835],
    [0.83, 0.52, 0.87],
]

# 3. Create DataFrames
df_stagnant = pd.DataFrame(
    stagnant, index=names, columns=["Dec", "Jan", "Feb"]
)
df_flowing = pd.DataFrame(flowing, index=names, columns=["Dec", "Jan", "Feb"])
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# 4. Display Raw Data Tables
print("=== SEDIMENT FROM STAGNANT WATER ===")
display(df_stagnant)

print("\n=== SEDIMENT FROM FLOWING WATER ===")
display(df_flowing)
print("\n" + "=" * 50 + "\n")


# 5. PCA Processing Function
def analyze_pca(df, title):
  X = StandardScaler().fit_transform(df)
  pca = PCA()
  scores = pca.fit_transform(X)

  scores_df = pd.DataFrame(scores, index=names, columns=["PC1", "PC2", "PC3"])
  loadings = pd.DataFrame(
      pca.components_.T, index=["Dec", "Jan", "Feb"], columns=["PC1", "PC2", "PC3"]
  )
  var_ratio = (pca.explained_variance_ratio_ * 100).round(1)

  print(f"--- {title} PCA RESULTS ---")
  print(f"Explained Variance Ratio (%): {var_ratio}")
  print("\nLoadings:")
  display(loadings.round(2))
  print("\n" + "=" * 40 + "\n")
  return scores_df, var_ratio


# Run PCA for both
scores_stagnant, var_stag = analyze_pca(df_stagnant, "STAGNANT WATER")
scores_flowing, var_flow = analyze_pca(df_flowing, "FLOWING WATER")

# 6. Seasonal & Congener Behaviour Summaries
print("=== STAGNANT WATER SEASONAL & CONGENER SUMMARY ===")
stag_means = df_stagnant.mean().round(2)
stag_totals = df_stagnant.sum().round(2)
stag_summary = pd.DataFrame(
    {"Mean Concentration": stag_means, "Total Accumulation": stag_totals}
)
display(stag_summary)
print(f"• December Highs (Stagnant): {df_stagnant['Dec'].nlargest(3).to_dict()}")
print(f"• January Highs (Stagnant):  {df_stagnant['Jan'].nlargest(3).to_dict()}")
print(f"• February Highs (Stagnant): {df_stagnant['Feb'].nlargest(3).to_dict()}")

print("\n" + "=" * 50 + "\n")

print("=== FLOWING WATER SEASONAL & CONGENER SUMMARY ===")
flow_means = df_flowing.mean().round(2)
flow_totals = df_flowing.sum().round(2)
flow_summary = pd.DataFrame(
    {"Mean Concentration": flow_means, "Total Accumulation": flow_totals}
)
display(flow_summary)
print(f"• December Highs (Flowing): {df_flowing['Dec'].nlargest(3).to_dict()}")
print(f"• January Highs (Flowing):  {df_flowing['Jan'].nlargest(3).to_dict()}")
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print(f"• February Highs (Flowing): {df_flowing['Feb'].nlargest(3).to_dict()}")
print("\n" + "=" * 50 + "\n")

# 7. Side-by-Side PCA Scatter Plots
fig, axes = plt.subplots(1, 2, figsize=(16, 7))
plot_data = [
    (scores_stagnant, var_stag, "Stagnant Water Sediment"),
    (scores_flowing, var_flow, "Flowing Water Sediment"),
]

for ax, (scores_df, explained, title) in zip(axes, plot_data):
  ax.scatter(scores_df["PC1"], scores_df["PC2"], color="teal", alpha=0.8)
  for n in names:
    ax.annotate(
        n.replace("PCB-", ""),
        (scores_df.loc[n, "PC1"], scores_df.loc[n, "PC2"]),
        fontsize=8,
        alpha=0.85,
    )

  ax.axhline(0, color="grey", lw=0.5, linestyle="--")
  ax.axvline(0, color="grey", lw=0.5, linestyle="--")
  ax.set_xlabel(f"PC1 ({explained[0]}%)", fontsize=10)
  ax.set_ylabel(f"PC2 ({explained[1]}%)", fontsize=10)
  ax.set_title(title, fontsize=12, fontweight="bold")

plt.tight_layout()
plt.savefig("pca_plot.png", dpi=300)
plt.show()
```

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=== SEDIMENT FROM STAGNANT WATER ===
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	Dec	Jan	Feb
PCB-8	2.745	0.570	2.505
PCB-18	10.830	0.755	1.080
PCB-28	3.950	0.730	0.570
PCB-44	19.425	0.820	1.005
PCB-52	3.220	0.515	1.695
PCB-66	10.770	1.245	0.515
PCB-77	0.635	1.310	0.555
PCB-81	8.250	0.680	0.540
PCB-101	5.555	0.550	1.290
PCB-105	10.640	0.770	0.625
PCB-114	4.700	0.635	0.720
PCB-118	2.810	1.535	3.235
PCB-123	4.085	0.395	2.230
PCB-126	6.440	1.525	1.550
PCB-128	4.460	0.745	1.965
PCB-138	2.390	0.405	0.815
PCB-153	9.005	0.900	1.570
PCB-156	5.875	1.010	0.510
PCB-157	4.255	1.375	1.985
PCB-167	23.345	0.510	1.400
PCB-169	4.160	0.565	1.090
PCB-170	6.600	0.710	2.240
PCB-180	7.570	2.100	2.005
PCB-187	4.780	0.850	1.090
PCB-189	7.920	0.280	0.775
PCB-195	12.935	0.520	1.330
PCB-206	7.340	0.630	1.015
PCB-209	17.030	0.715	1.880

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=== SEDIMENT FROM FLOWING WATER ===
```

	Dec	Jan	Feb
PCB-8	3.350	0.110	0.620
PCB-18	2.350	0.310	0.325
PCB-28	1.350	0.800	0.820
PCB-44	4.665	0.820	0.455
PCB-52	3.680	0.535	0.275
PCB-66	2.940	0.340	1.355
PCB-77	0.315	0.620	0.225
PCB-81	2.025	0.350	0.355
PCB-101	0.665	0.465	0.795
PCB-105	2.705	0.430	1.190
PCB-114	0.920	0.305	0.825
PCB-118	3.835	0.365	0.295
PCB-123	0.600	1.130	1.230
PCB-126	0.635	0.310	0.825
PCB-128	0.710	0.805	0.410
PCB-138	1.200	0.940	0.850
PCB-153	3.145	0.770	1.620
PCB-156	1.660	0.605	0.615
PCB-157	1.245	0.620	0.735
PCB-167	2.095	0.530	0.995
PCB-169	1.710	0.660	2.365
PCB-170	3.875	0.720	1.135
PCB-180	2.960	0.305	1.000
PCB-187	1.735	0.530	2.010
PCB-189	4.985	0.600	0.620
PCB-195	3.900	0.465	0.995
PCB-206	3.135	0.745	0.835
PCB-209	0.830	0.520	0.870

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--- STAGNANT WATER PCA RESULTS ---
Explained Variance Ratio (%): [44.3 30.1 25.6]

Loadings:
```

	PC1	PC2	PC3
Dec	-0.48	0.88	0.02
Jan	0.62	0.33	0.71
Feb	0.62	0.35	-0.70

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--- FLOWING WATER PCA RESULTS ---
Explained Variance Ratio (%): [42.  31.1 26.9]

Loadings:
```

	PC1	PC2	PC3
Dec	-0.52	0.75	0.40
Jan	0.65	0.04	0.76
Feb	0.55	0.66	-0.51

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=== STAGNANT WATER SEASONAL & CONGENER SUMMARY ===
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	Mean Concentration	Total Accumulation
Dec	7.56	211.72
Jan	0.83	23.35
Feb	1.35	37.78

```
• December Highs (Stagnant): {'PCB-167': 23.345, 'PCB-44': 19.425, 'PCB-209': 17.03}
• January Highs (Stagnant):  {'PCB-180': 2.1, 'PCB-118': 1.535, 'PCB-126': 1.525}
• February Highs (Stagnant): {'PCB-118': 3.235, 'PCB-8': 2.505, 'PCB-170': 2.24}

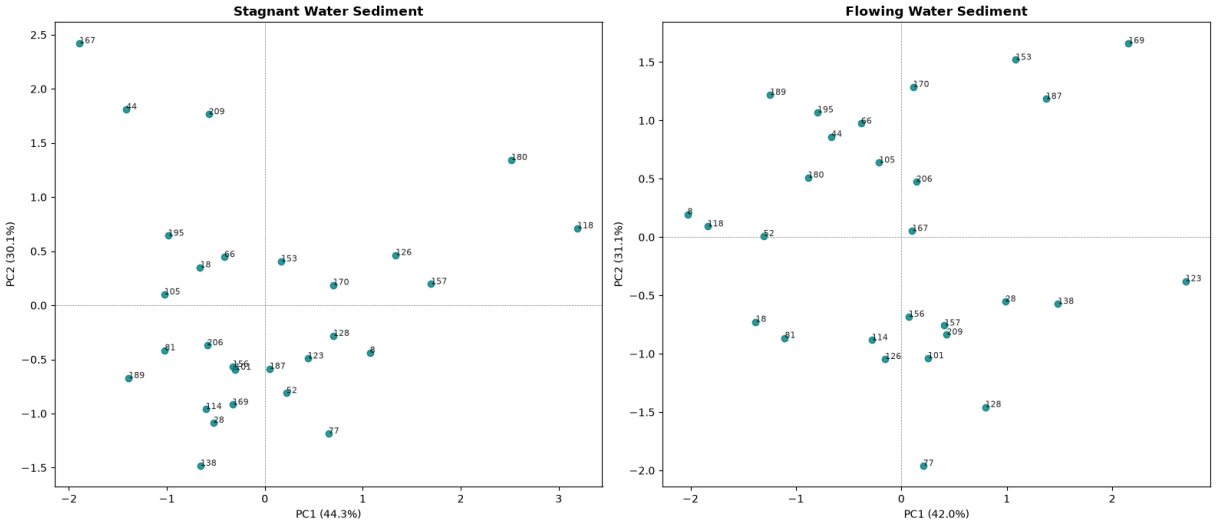
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=== FLOWING WATER SEASONAL & CONGENER SUMMARY ===
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	Mean Concentration	Total Accumulation
Dec	2.26	63.22
Jan	0.56	15.70
Feb	0.88	24.65

- December Highs (Flowing): {'PCB-189': 4.985, 'PCB-44': 4.665, 'PCB-195': 3.9}
- January Highs (Flowing): {'PCB-123': 1.13, 'PCB-138': 0.94, 'PCB-44': 0.82}
- February Highs (Flowing): {'PCB-169': 2.365, 'PCB-187': 2.01, 'PCB-153': 1.62}

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